# Sanaul Haque

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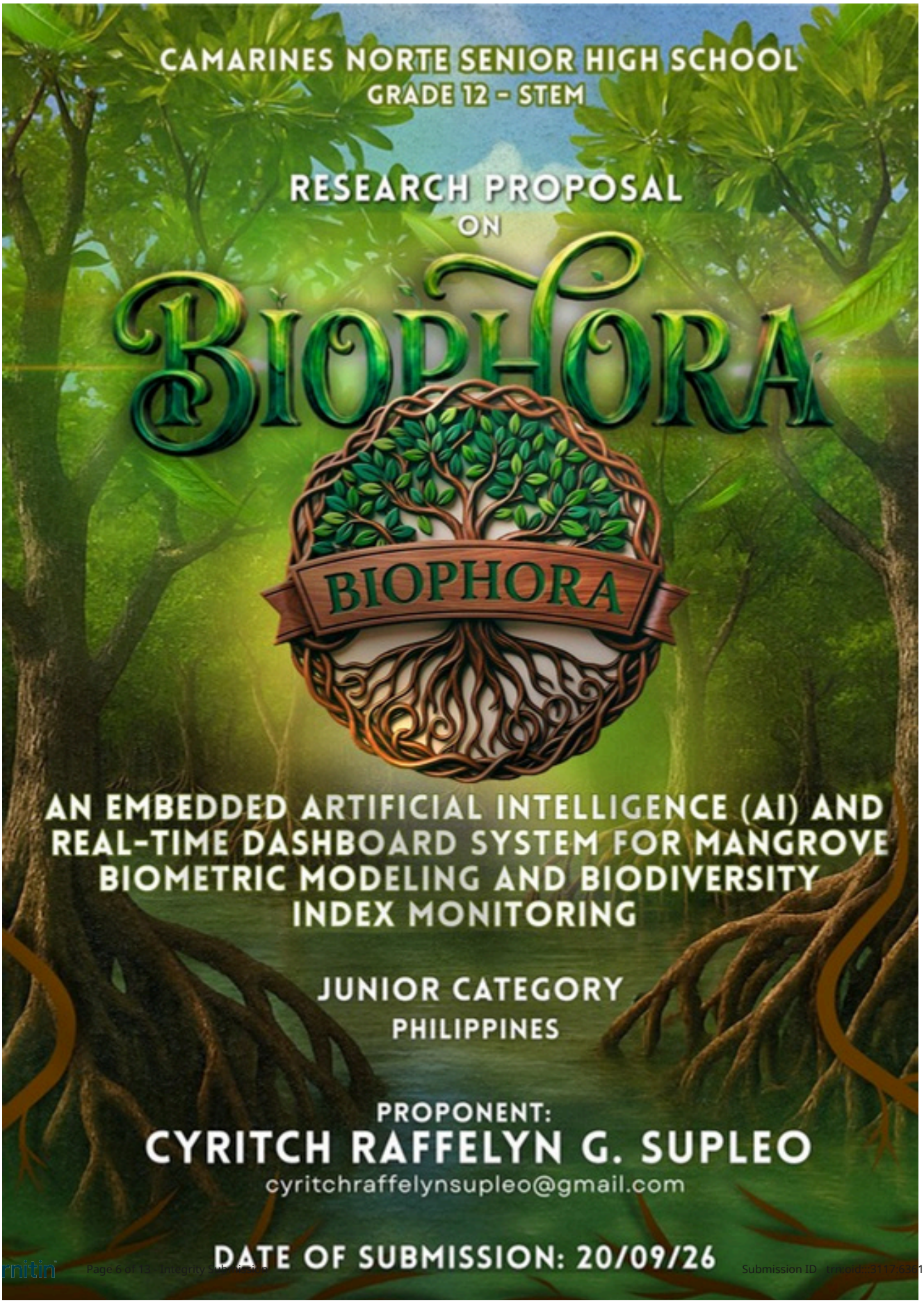
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CAMARINES NORTE SENIOR HIGH SCHOOL
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RESEARCH PROPOSAL
ON
BIOPHORA
BIOPHORA
AN EMBEDDED ARTIFICIAL INTELLIGENCE (AI) AND REAL-TIME DASHBOARD SYSTEM FOR MANGROVE BIOMETRIC MODELING AND BIODIVERSITY INDEX MONITORING
JUNIOR CATEGORY
PHILIPPINES
PROPONENT:
CYRITCH RAFFELYN G. SUPLEO
cyritchraffelynsupleo@gmail.com
DATE OF SUBMISSION: 20/09/26

Title: BIOPHORA: An Embedded Artificial Intelligence (AI) and Real-Time Dashboard System for Mangrove Biometric Modeling and Biodiversity Index Monitoring

**Research Problem**

Mangrove forests are among the most important natural ecosystems found in the world they can protect coastal regions from storms and prevent marine degradation, store Carbon, provide habitat to different species of flora and fauna and act as a great source of Natural Resources for coastal communities. However, continuous deforestation pollution, coastal developments, overexploitation and climate change still remain a huge issue for these ecosystems. The traditional way of mangrove monitoring consists of visual identification and measurement of vegetation, spatial data collection dredging coring and sediment elevation measurements biodiversity assessments. Despite generating valuable ecological data, these systems can be extremely labour intensive, time consuming and costly to operate in remote mangrove environments.

Development in certain emerging technologies such as embedded Artificial intelligence(AI), Internet of Things (IoT), remote sensing, and digital dashboards have enabled environmental monitoring through automated data collection, image analysis, spatial mapping and visualisation. On the other hand, only a few were found that can integrate all aspects such as: AI-assisted species identification and observable characteristic analysis (leaf and mangrove), biometric modelling, biodiversity index computation, ecosystem health assessment environmental Risk assessment, spatial mapping and dashboard visualization in a single field-oriented system. **Existing Literature**

Previous studies have explored different technologies for mangrove monitoring, including artificial intelligence (AI), computer vision (CV), remote sensing (RS), geospatial technologies, and the Internet of Things (IoT). Wäldchen and Mäder [1] demonstrated the use of computer vision for automated plant species identification, providing a basis for image-based recognition of mangrove species. Trecene et al. [2] applied fine-grained classification and deep learning techniques to identify mangrove species, while Cheng et al. [3] demonstrated the use of deep learning for mangrove species identification and biodiversity monitoring. Farhan et al. [4] also explored the use of convolutional neural networks for plant classification based on observable physical characteristics. Arias et al. [5] applied learning algorithms to support the automated monitoring and conservation of mangrove ecosystems. Remote sensing and geospatial technologies have likewise been used for mangrove mapping and ecosystem assessment. Pillodar et al. [6] used remote sensing images for mangrove mapping in the Philippines, while Pimple et al. [7] examined temporal changes in mangrove distribution. Wang et al. [8] combined UAV-LiDAR and Sentinel-2 imagery for mangrove species mapping, and Sunkur et al. [9] investigated the effectiveness of remote sensing for mangrove mapping and assessment. Zhao et al. [10] also used Sentinel-2 imagery for mangrove classification. Perito [11] documented mangrove species in Davao del Sur and demonstrated the use of geotagging, online data storage, and image-based leaf identification for visualizing mangrove distribution. These studies show that AI, image processing, remote sensing, and GIS can support more systematic and data-driven approaches to species identification, biodiversity assessment, and spatial monitoring.

Other studies have integrated sensors, IoT, image processing, machine learning, and digital platforms for environmental monitoring. Boonrat et al. [12] developed a wireless protocol using IoT sensors to collect and transmit environmental data for mangrove monitoring. Baynas and De Ocampo [13] designed a mangrove habitat monitoring system that combined sensor networks with machine vision and underwater image analysis. Lomeo and Singh [14] developed a cloud-based monitoring approach that combined deep learning and satellite imagery to examine mangrove distribution in Southeast Asia. In the Philippines, Zablan et al. [15] combined satellite imagery, GPS ground surveys, and machine learning to validate mangrove distribution maps. Dhawi et al. [16] developed an AI- and IoT-based dashboard for mangrove monitoring in India, while Zamani et al. [17] presented an intelligent biodiversity dashboard for the visualization and management of biodiversity data. These studies demonstrate how technology can improve the collection, processing, analysis, mapping, and visualization of ecological information.

Overall, the reviewed studies demonstrate that AI and computer vision can support species identification, remote sensing and geospatial technologies can assist in mangrove mapping, IoT and sensor systems can facilitate environmental monitoring, and dashboards can improve the visualization of ecological data. However, most of these approaches focus on specific monitoring functions or use separate platforms and data sources. Based on the reviewed literature, there remains a need for an integrated in-field system that combines AI-based species identification, observable mangrove characteristics, biometric data, biodiversity index computation, environmental threat assessment, GPS-based spatial mapping, and dashboard visualization. BIOPHORA addresses this gap by proposing these functions within a single integrated mangrove monitoring system.

**Research Questions**

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This study aims to develop and evaluate BIOPHORA as an integrated system for monitoring and assessing mangrove ecosystems. Specifically, it seeks to answer the following questions:

1. How effective is BIOPHORA in monitoring mangrove ecosystems in terms of:
    a. Mangrove Biological Characteristics
        a.1. AI-based Mangrove Species Identification
        a.2. Leaf Characteristics Analysis
    b. Ecosystem Health and Threat Status
        b.1. Mangrove Health Status Classification
        b.2. Environmental Risk Detection
    c. Phytosociological Parameters
        c.1. Species Richness Assessment
        c.2. Species Distribution Monitoring
        c.3. Biodiversity Index Computation
    d. System Functional Features
        d.1. Biometric Data Presentation
        d.2. Biodiversity Index Visualization
        d.3. Spatial Mapping of Mangrove Populations
        d.4. Real-Time Dashboard Monitoring and Reporting?

2. Is there a significant difference in the consistency and accuracy of the system's results across different study areas?
3. What is the level of acceptability of BIOPHORA based on selected ISO/IEC 25010 software quality characteristics, particularly:
   a. Accuracy;
   b. Reliability;
   c. Functional Suitability; and
   d. Performance Efficiency

**Methodology**

This study adopts a quantitative method which will help to assess the performance and the viability of the BIOPHORA system.

Quantitative Component:

- An experimental design will be employed to assess species identification, biometric modeling, biodiversity monitoring, ecosystem health observation and spatial mapping of BIOPHORA.
- The four sites in the mangrove estimation, which include: Bagasbas Awitan Gubat and Mambalite will be subject to sampling.
- Comparison of system results (obtained from the different study areas) will be made using One-Way ANOVA; Species richness and distribution will be determined from the verified species observations collected in each study area.
- Simpson's Diversity Index will then be calculated from the species abundance data to quantify biodiversity.
- Weighted Mean shall be used to assess the system's acceptability criteria based on the selected characteristics of ISO/IEC 25010.

**1. Hardware Development:**

The Raspberry Pi 5 will serve as the processing unit, with a 2K autofocus camera and an external GPS module to collect geospatial data in handheld BIOPHORA. The flow of data will go through ESP32 and the internet module to the mobile app and dashboard. The information and status of the system will be displayed in a 4" Raspberry Pi display. 64 GB SD card–this is where the system data will be stored. The device consists of 3 Ah LiFePO4 batteries, a 100A (120A MAX) 4S Daly BMS, buck converter, fuse box and LiFePO4 charger. A rugged 3D printed enclosure will hold all components firmly in place.

**2. Software Development:**

The data in the form of images and details of identified mangrove species will be collected in a dataset. The dataset will first be pre-processed and this will help in training and validating the AI model for species identification. Once validated, this model will be deployed on Raspberry Pi to perform image classification. It also applies observable leaf and mangrove characteristics in health and environmental risk detection through a set of quantitative criteria derived from scientific literature and verified by professionals from environmental or forestry fields. Every

actual observation will therefore have GPS coordinates marking exactly where it was located. The BIOPHORA device will include an embedded internet communication module that will transmit the data collected to the mobile application and dashboard. Field measurements (such as biometric metrics, biodiversity indices, species identification as well) localization (where it was found), and more will be displayed to the user in a dashboard for both: distributions of species; and/or health or risk. This also means displaying data on the dashboard with minimal latency using real-time monitoring when there is a connection to the network. If there is no data connection available, the information will be stored locally on the device and synchronized when re-establishing a valid connection. In addition, it will record the transmission time and subsequently create monitoring reports to be submitted to authorized personnel in the Department of Environment and Natural Resources (DENR) & partner agencies.

**3. Testing and Evaluation:**

The BIOPHORA system is tested both in bench testing and in field trials. Bench testing will test each component including the camera and image acquisition, AI species identification, biometric processing, GPS acquisition, biodiversity calculations, data storage and dashboard communication. The system will be field tested in the chosen four mangrove areas where species identification, biometric measurements, coordinates and biodiversity data, its working fitness for classifying ecosystem health or environmental risk detection as well as transmitting dashboard data under actual field conditions can be evaluated. The outputs of BIOPHORA will then be compared with reference or expert-verified measurements and classifications to evaluate whether the system reliably measures biological or chemical parameters as intended.

**4. Acceptability Rating:**

BIOPHORA will be evaluated by experts with a four-point Likert-type scale based on the ISO/IEC 25010 software quality properties, most notably Functional Suitability, Reliability having Weightage to Performance Efficiency. The accuracy is also a major metric of performance regarding how the system outputs are judged, in particular, species identification correctness, biometric measurements and biodiversity indices as well as other results generated. We can say that Functional Suitability will include correctness and completeness of system functions and outputs whereas Reliability and Performance Efficiency will evaluate consistency and responsiveness of the system in operation.

**5. Performance Evaluation and Recommendations:**

The performance of all the distinct modules of the system should be evaluated using species identification accuracy, accuracy of every biometric measurement, GPS placement accuracy, biodiversity index compensating accuracy, healthy ecosystem compensating density as well as user acceptability. Quantitative data collected during fieldwork will be statistically analyzed, dependent on the study sites. Quantitative results will be presented descriptively (e.g. mean and

error measures as appropriate). For the study areas where assumptions for the test are fulfilled, One-Way ANOVA will be performed to check if there are significant differences in mean measurement errors of BIOPHORA among the four study areas. We will assess species-identification performance via an appropriate classification metric and system acceptance based on a weighted mean of expert evaluation response. Study results will be compared, as appropriate, to determine whether the observed number of outliers using accuracy and precision of outputs of the system between various study sites are significantly different from one another. Potential recommendations on calibration or other aspects such as species classification through AI, biometric measurements, GPS mapping, collected biodiversity data and system dashboard to enhance the reliability and 'useability' of BIOPHORA and field application for mangrove biodiversity monitoring.

Ethical Considerations:

- All necessary institutional and site permissions will be secured before field activities.
- Appropriate permissions from relevant environmental authorities and local stakeholders will be obtained before collecting ecological and geospatial data.
- If experts or other individuals participate in system evaluation, informed consent will be obtained, and their responses will be treated confidentially and anonymously.
- Field activities will follow appropriate environmental and safety protocols to minimize disturbance to mangrove ecosystems and reduce risks to the researcher.

**Research Topic**

Research aims to create an integrated technological platform for mangrove ecosystems monitoring and assessment. BIOPHORA gathers together Embedded Artificial Intelligence, mangrove biometry, biodiversity index monitoring, environmental risk detection, spatial mapping and real-time data visualization on a single system.

This research is particularly relevant to mangrove conservation because the ecological information generated by the system may support researchers, environmental managers, conservationists, and local government units in monitoring and managing mangrove ecosystems.

**Quality of Writing**

The proposal employs clear, academically appropriate, technically correct language to support the use of a multi-disciplinary target audience. The terminology employed is necessary and used to support clarity of the proposal. Standard research conventions are followed and the terminology, use of grammar, citation style and the format are consistent and correct for submission to IARCO.

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